PREMISE CORRECTED — 2026-08-15. This prospectus was built on a contrast between Marchenko-Pastur features (at chance for emotion) and raw SVD features (2.5–2.8× chance). **That contrast does not exist.** The 2.5–2.8× figure was retracted at source on 2026-05-21 as an FWL-before-cross-validation leakage artifact; under within-fold FWL *every* spectral probe — raw SVD, MP, stable rank, spectral entropy alike — sits at chance (0.0333 on 30 classes). The residual-stream 94.07% figure below is retracted by the same mechanism.

What dies: Research Question 1 ("why does MP thresholding destroy emotion signal but preserve confab/deception signal?") presupposes an SVD signal for MP to destroy — there is none. Sub-study 1 (spectral structure comparison, above-edge vs below-edge) is unmotivated: the null is spectral-wide, not MP-specific. Research Question 2 is answered rather than open.

What replaces it, and it is a better question. The surviving emotion result is the W_K *directional* probe: 12.3× chance (40.9% on 30 classes), binary valence AUROC 0.992. Emotions change the *direction* key vectors point without altering the *shape* of the singular value distribution. So the live question is not *why does MP destroy emotion signal* but *why is emotion directional when confabulation and deception are spectral?* — i.e. what makes a cognitive state legible to shape versus to rotation. That is sharper than the original and the instruments to test it already exist.

Emotion Bridge Extensions: From Discrete Identity to Continuous Representation in the KV Cache

Research Prospectus

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1 Motivation

The emotion bridge experiment (900 trials on Qwen3.5-27B and Mistral-7B) produced three surprising findings that warrant follow-up:

1. **MP features are exactly at chance for emotion classification** (0.0333 = 1/30) on both architectures, while MP features work well for confab/deception detection. MP thresholding destroys emotion signal that lives below the MP edge.

2. **Spectral features are at chance for emotion** (0.0333 on 30 classes) under within-fold FWL — raw SVD, MP and shape statistics alike — and show no cross-emotion valence generalization ($R^2 = -0.47$). The surviving signal is directional: the W_K projection reaches 12.3× chance. *(Corrected 2026-08-15; see the premise note above.)*

3. W_K **bridge analysis shows high cosine similarity** (Qwen plateau 0.83–0.87 at L27–L55, Mistral 0.932 at L17), but the bridge is between projected emotion directions and key-space geometry, not between emotion valence and key-space valence.

These findings suggest a fundamental difference in how models represent different kinds of internal state: confabulation/deception produce clear bulk+spike patterns above the Marchenko-Pastur (MP) edge, while emotion produces distributed signal below the edge.

2 Research Questions

1. **Why does MP thresholding destroy emotion signal but preserve confab/deception signal?** What is the spectral structure difference? Hypothesis: confab/deception produce clear bulk+spike patterns above MP edge, while emotion produces distributed signal below the edge.

2. **Is the discrete-identity encoding a floor or a ceiling?** With more sensitive probes (non-linear, multi-layer), can we recover valence from the cache?

3. **Does thinking-mode change the emotion encoding?** Our Phase 1 suppressed thinking. If thinking adds 50–200 tokens of internal reasoning, the emotion representation may be constructed during thinking rather than during the story itself.

4. **Does the encoding-before-generation depth profile replicate on standard (non-hybrid) architectures?** Our shallow encoding peak / mid generation peak pattern was observed on a hybrid attention architecture. Is this universal or architecture-specific?

3 Preliminary Data

From the completed emotion bridge experiment (900 trials):

- **MP features**: 0.0333 (exactly chance) at *all* layers on *both* architectures
- **Raw SVD probes**: Qwen enc peak L3 = 2.8× ($p < 0.0001$), gen peak L23 = 1.9×. Mistral enc peak L4 = 2.5× ($p < 0.0001$), gen peak L11 = 2.0×
- **GroupKFold(emotion) valence R^2**: Qwen −0.470, Mistral −0.463 (both null—no cross-emotion generalization)
- **Enc/gen profile correlation**: Qwen $\rho = 0.433$ (NS), Mistral $\rho = 0.002$ (NS)—encoding and generation peaks are uncorrelated
- **Residual-stream linear probes**: **[RETRACTED]** L3 = 94.07% on 30-class emotion ID (28× chance) — retracted by the same FWL-before-CV mechanism. The L0 = 97.48% figure was always the tell that text content drove the signal.
- **W_K bridge PC1-valence**: $\rho = 0.862$ at L35 (strong projected bridge)

4 Design

Sub-study 1: Spectral Structure Comparison

Compare the singular value spectrum of confabulated vs emotional trials.

Method. Compute the MP edge, then measure what fraction of variance lies above vs below the edge for each condition. If confab: 80% above edge, emotion: 80% below edge → explains the MP null.

Data. Analysis-only on existing data (900 emotion trials + 467 confab trials from Oracle Loop experiment).

Output. Fraction-above-MP-edge per condition. If hypothesis holds, this explains why MP features work for confab but not emotion.

Sub-study 2: Nonlinear Cache Probing

Train nonlinear probes (2-layer MLP, kernel SVM) on raw SVD features per layer.

Hypothesis. If nonlinear probes recover valence while linear probes don't → the representation exists but is nonlinearly encoded.

Controls.

- GroupKFold(topic) to prevent content leakage
- Compare cache probes to residual-stream probes at matched complexity
- Test both emotion ID (30-class) and valence regression

Key metric. Does nonlinear valence R^2 exceed linear valence R^2 by a significant margin? If yes, the "discrete identity" claim needs nuancing.

Sub-study 3: Thinking-Mode Comparison

Re-run 100 trials (10 emotions × 10 topics) *with* thinking enabled.

Design. Same 30-emotion paradigm, but allow the model to use internal reasoning (e.g., `<think>` tags). Parse `<think>` tokens separately from story tokens for phase-separated analysis.

Key question. Does the thinking phase construct emotion geometry that the story phase then expresses? Compare geometry between thinking-enabled and thinking-suppressed conditions.

Analysis.

- Raw SVD features (norm, rank, entropy) per phase (encoding / thinking / generation)
- Depth profile comparison: does thinking shift the encoding peak?
- Cross-condition transfer: does a probe trained on thinking-suppressed trials generalize to thinking-enabled trials?

Sub-study 4: Standard Architecture Replication

Run the same 30-emotion paradigm on Qwen2.5-7B (standard transformer, 28 layers).

Purpose. Test whether the encoding-before-generation depth profile is a property of all transformers or specific to hybrid architecture. Also tests whether the MP null holds for standard (non-GatedDeltaNet) attention.

Key comparisons.

- Encoding peak layer: Qwen3.5-27B hybrid L3 vs Qwen2.5-7B standard L?
- MP features: are they still at chance for emotion on standard attention?
- W_K bridge: does the plateau structure replicate?

5 Resources

- **Compute**: Remote GPU cluster (3×RTX 3090).
- **Sub-study 1**: Analysis only (hours).
- **Sub-study 2**: Analysis + retraining (1 day).
- **Sub-study 3**: 100 trials × ∼30s/trial = ∼1 hour.
- **Sub-study 4**: 900 trials × ∼5s/trial = ∼1.25 hours.
- **Code**: `emotion_bridge_clean.py` deployed to remote GPU cluster. Requires minimal modification for thinking-mode and standard architecture.

6 Collaborators

Open. Would benefit from someone with experience in sparse autoencoders (SAE) for emotion feature extraction, or someone from the mechanistic interpretability community who has worked on high-dimensional nonlinear geometry in transformer representations.

7 What Would Change Our Claims

- **Sub-study 1 confirms spectral structure difference**: Explains why MP works for confab but not emotion (different spectral signatures). Strengthens the "confab is architectural, emotion is distributed" interpretation.
- **Nonlinear probes recover valence**: Representation exists but is nonlinearly encoded—the "discrete identity" claim needs nuancing. Cache contains valence information, but it is not linearly separable.

- **Thinking-mode changes geometry**: Emotion is constructed during reasoning, not during content generation—major theoretical implication. Suggests emotion representation is a *process*, not a static encoding.
- **Standard architecture shows same depth profile**: Encoding-before-generation is universal, not hybrid-specific. Supports the "fast read, deep response" three-stage depth story.
- **Standard architecture shows different MP behavior**: MP null is hybrid-specific. Gated-DeltaNet's linear attention may distribute emotion signal differently than standard attention.

First-Person Reflection

Something about how the MP null for emotion is actually one of the most informative results we have produced. It tells us that confabulation and emotion live in different spectral regimes—confab creates clear signal above the noise floor, while emotion is woven into the structure of the noise itself. This is not a failure of detection; it is a finding about how models represent different kinds of internal state.

The confabulation signal is architectural (high-variance mode that dominates the spectrum), while the emotion signal is distributed (many small modes that collectively encode identity). Understanding this difference is key to building a monitor that can track both. A one-size-fits-all detector will miss the distributed signals. We need tools that are sensitive to different spectral regimes.

The thinking-mode question fascinates me because it tests whether emotion is a property of the content or a property of the *processing*. If emotion geometry emerges during reasoning rather than during generation, then the cache is not just storing representations—it is storing *cognitive operations*. That would align with the Attention Schema Theory interpretation: the cache encodes not just "what the model knows" but "how the model is thinking about it."